Fiscal Policy with Heterogeneous Agents I: Government Debt and Transitions

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Plan of the Topic

1. The government budget: identities, debt dynamics, data.
2. The benchmark: **Ricardian equivalence**—and the assumptions doing the work.
3. Government debt in the Aiyagari economy: liquidity vs. crowding out, with a computed experiment.
4. **Aiyagari–McGrattan (1998)**: the optimum quantity of debt.
5. **Transition dynamics**: the algorithm.
6. **Domeij–Heathcote (2004)**: why transitions and distribution overturn steady-state welfare rankings.

The Flow Budget Identity

In each period, with debt B_t (one-period, riskless), spending G_t, transfers Tr_t, and tax revenue T_t:

$$B_{t+1} = (1 + r_t)\, B_t + G_t + Tr_t - T_t.$$

- Define the **primary surplus** $S_t \equiv T_t - G_t - Tr_t$: the budget net of interest.
- Then $B_{t+1} - B_t = r_t B_t - S_t$: debt grows by interest minus the primary surplus.
- Accounting, not economics—but every model in this topic must respect it, period by period.

Debt Dynamics: the $r - g$ Arithmetic

Divide by output Y_t, growing at rate g; lowercase = ratios:

$$b_{t+1} = \frac{1+r}{1+g}\, b_t - s_t \qquad \Longrightarrow \qquad \Delta b \approx (r-g)\, b - s.$$

- $r > g$: the debt ratio snowballs without primary surpluses; stabilizing b requires $s = (r-g)b$.
- $r < g$: the ratio melts away on its own; deficits can be permanent ("free lunch" debates).
- In the models of this topic, r is an *equilibrium object that depends on b*—the whole point of the HA analysis is that issuing debt moves the interest rate you must pay on it.

Solvency: No-Ponzi and the Intertemporal Budget

Iterating the flow identity forward and imposing $\lim_{T\to\infty} B_T / \prod(1+r) = 0$ (no Ponzi schemes):

$$(1+r_0)\,B_0 \;=\; \sum_{t=0}^{\infty} \frac{S_t}{\prod_{j=1}^{t}(1+r_j)}.$$

- Outstanding debt must equal the present value of future primary surpluses.
- Debt is therefore *deferred taxation*: the question is never whether to pay, only **when** and **who**.
- "When and who" is precisely what representative-agent complete-markets models make irrelevant—next part—and what heterogeneous-agent models make central.

A Worked Budget Line

Illustration with round numbers: $b = 1.0$ (debt equal to annual output), $r = 4\%$, $g = 2\%$.

- Stabilizing the ratio requires a permanent primary surplus of $(r - g)\, b = 2\%$ of output.
- If instead $r = 1.5\%$ (below g): the government can run a permanent primary *deficit* of 0.5% of output with a stable ratio.
- Conclusion: a 250-basis-point movement in r swings the required fiscal stance by 2.5% of output at this debt level. What pins r down is importanimportantt.

Where the U.S. Sits (Approximate)

- Federal debt held by the public: around 100% of GDP.
- Net interest outlays: roughly 3% of GDP and rising with refinancing at higher rates.
- Real safe rates: negative to near-zero through the 2010s, positive again since—the r vs. g sign has flipped within a decade.

The theory questions this topic answers: what does public debt *do* in an economy where households self-insure, what is the *optimal* quantity, and how do we evaluate a change—not just across steady states, but along the path?

Two Opposite Intuitions

- **Crowding out** (undergraduate intuition): government borrowing competes with firms for savings, raises r, lowers K—debt is costly.
- **Liquidity provision** (incomplete-markets intuition): households facing uninsurable risk are *short* of safe assets to buffer with; public debt supplies exactly that asset—debt is valuable.
- Both are right; the question is quantitative, and the answer runs through the asset-market diagram of the Aiyagari economy.
- First, the benchmark in which *neither* operates.

The Statement

Ricardian equivalence. In a representative-agent economy with complete markets and lump-sum taxes, the timing of taxes— and hence the path of government debt—is irrelevant: for a given path of government spending $\{G_t\}$, every financing scheme supports the *same* equilibrium allocation and prices.

- Debt-financed tax cuts do not stimulate consumption; they are saved.
- Government debt is not net wealth (Barro 1974).

Environment for the Proof

- Representative household, endowment or production economy, complete markets, discount factor β.
- Government: spending path $\{G_t\}$, **lump-sum** taxes $\{T_t\}$, debt B_t, flow budget as before.
- Household holds debt and capital; both pay return r_t (no arbitrage).

Strategy: show that taxes enter the household's problem only through their *present value*, which the government's own solvency condition pins to fundamentals.

Proof, Step 1: the Household's Lifetime Budget

Iterate the household's flow budget forward (with its own no-Ponzi condition); discounting by $q_t \equiv \prod_{j \le t}(1 + r_j)^{-1}$:

$$\sum_t q_t \, c_t \;=\; \underbrace{(1 + r_0)(a_0 + B_0)}_{\text{initial wealth}} \;+\; \sum_t q_t \, y_t \;-\; \sum_t q_t \, T_t .$$

- Only the **present value** of taxes appears; their timing has vanished already.
- Debt appears twice: as an asset in initial wealth, and implicitly through future T_t.

Proof, Step 2: Substitute Government Solvency

The government's intertemporal budget:

$$\sum_t q_t T_t = (1 + r_0) B_0 + \sum_t q_t G_t.$$

Substitute into the household's lifetime budget:

$$\sum_t q_t c_t = (1 + r_0) a_0 + \sum_t q_t y_t - \sum_t q_t G_t.$$

- B_0 and the entire tax *path* cancel out: the household's opportunity set depends only on $\{G_t\}$.
- Same opportunity set $\Rightarrow$ same choices $\Rightarrow$ same equilibrium. $\square$
- The bond the household "owns" is exactly offset by the taxes it "owes": debt is not net wealth.

The Assumptions Doing the Work

Each cancellation used one assumption; each fails somewhere real:

1. **Lump-sum taxes.** Real taxes distort margins
2. **No borrowing constraints / complete markets.** A constrained household cannot undo a tax deferral by borrowing against it—timing has real effects.
3. **Infinite horizon** (or perfect intergenerational altruism, Barro 1974). Finite lives without bequest motives break the offset.
4. **Same discounting for household and government** (one interest rate; no risk premia on either side).

Which Failures Matter Quantitatively?

- The HA literature's answer: assumptions 1 and 2, jointly. With uninsurable risk and borrowing limits, a sizable mass of households is at or near the constraint, with high MPCs—tax timing moves *their* consumption one for one.
- Heathcote (2005): in a calibrated incomplete-markets economy, a debt-financed tax cut raises aggregate consumption substantially—quantitatively large departures from equivalence, driven by exactly the constrained households.
- So we do not *assume* Ricardian equivalence away; we break it with the same frictions the whole course runs on, and then measure.

What Survives

- The *logic* survives as an accounting discipline: debt is still deferred taxation; the intertemporal budget still binds; "who pays, and when" is still the whole question.
- What changes in HA economies: the answer to "who" differs across the wealth distribution, the answer to "when" moves real allocations, and debt performs a *function*— it is an asset someone wants to hold.
- That function is where we go next.

The Environment

The heterogeneous labor markets models, plus a government:

- Households: CRRA utility, idiosyncratic productivity e , borrowing limit, save in a single riskless asset.
- Firms: $Y = K^{\theta}L^{1-\theta}$, depreciation δ.
- Government: debt B (constant, steady state), services interest rB with a proportional labor tax τ: $\tau wL = rB$.
- Asset-market clearing is the new line:

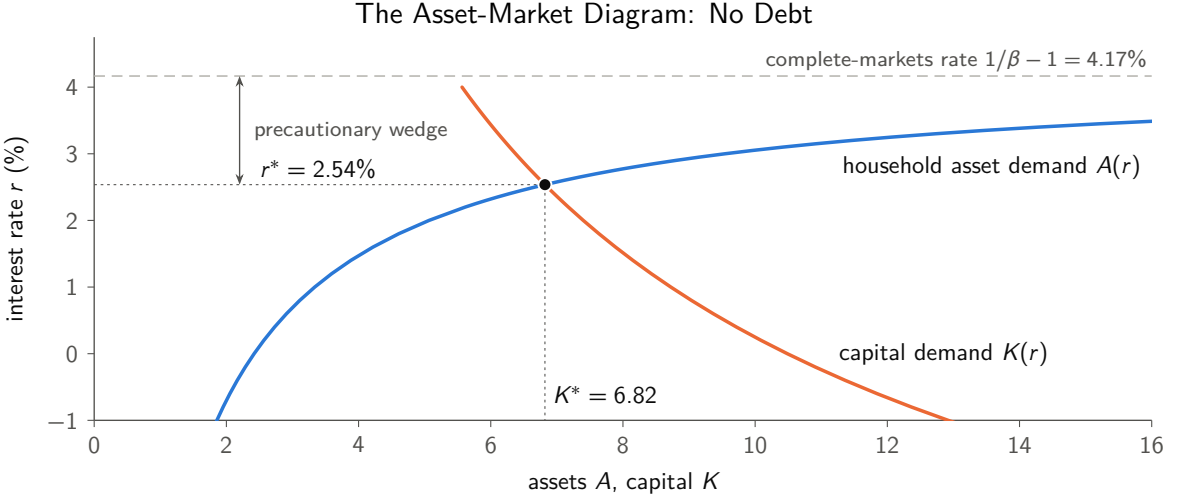
$$\underbrace{A(r)}_{\text{household asset demand}} = K(r) + B.$$

The Asset-Market Diagram, Recalled

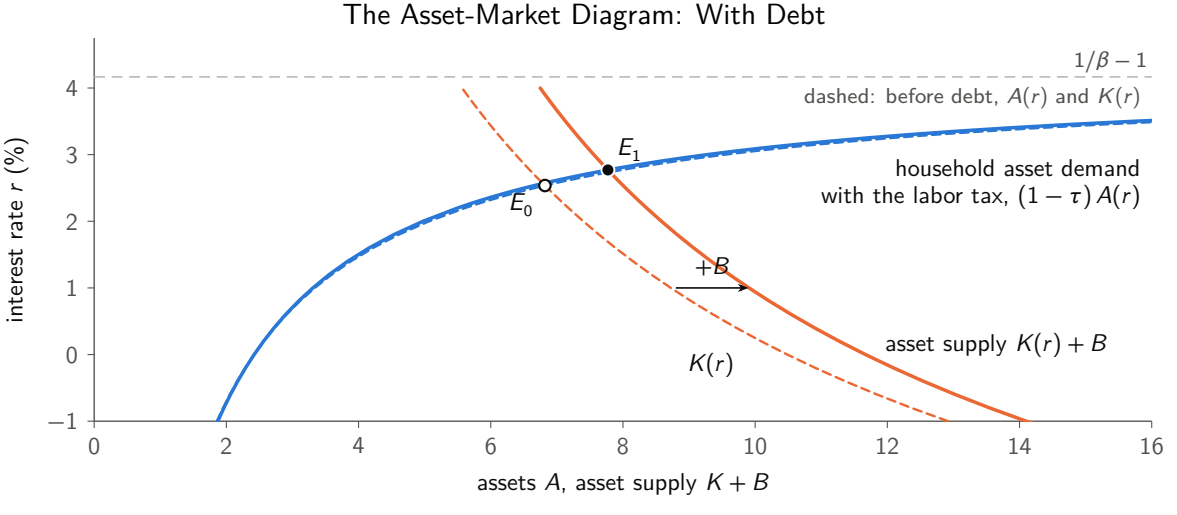
From 8002, the two curves in (A, r) space:

- **Household asset demand** $A(r)$: upward-sloping, and—the Aiyagari signature—exploding as $r \to 1/\beta - 1$: near the complete-markets rate, precautionary savers want unbounded buffers.
- **Capital demand** $K(r)$ from the firm's FOC: downward-sloping.
- Equilibrium without debt: $A(r^*) = K(r^*)$, with $r^* < 1/\beta - 1$—the precautionary wedge.

Government debt shifts the *supply of assets* to $K(r) + B$: households can now hold paper as well as capital.



Calibration of the experiment below. Demand explodes as $r \to 1/\beta - 1$, so $r^* = 2.54\%$ lies below it.



Supply shifts right by B (here $B/Y = 0.6$); the labor tax that services it scales demand by $(1 - \tau)$.

What Debt Does: Comparative Statics

Raise B. The asset-supply schedule shifts right; equilibrium moves *up* the household demand curve:

- r **rises**: paper competes with capital for the household's portfolio.
- K **falls**: crowding out—but only by the amount household wealth fails to expand.
- Total private wealth A **rises**: households willingly absorb most of the new paper, because at a higher r, self-insurance is cheaper.
- w falls with K/L; the tax τ rises to service rB—both costs land on labor income.

How much of each? Numbers, next.

A Computed Experiment: Setup

Annual calibration, deliberately standard:

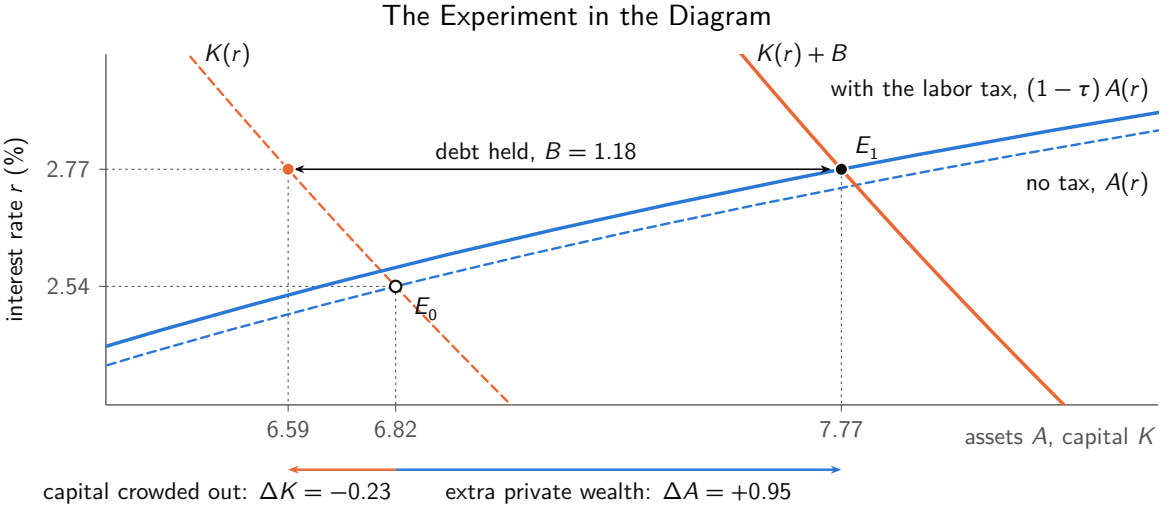
Object	**Value**	**Object**	**Value**
β	0.96	μ (CRRA)	2
θ	0.36	δ	0.08
ρ_e	0.90	σ_e	0.20
borrowing limit	0	tax	labor tax balances $\tau wL = rB$

Experiment: steady state at $B/Y = 0$ versus $B/Y = 0.6$.

A Computed Experiment

	$B/Y = 0$	$B/Y = 0.6$
interest rate r	2.54%	2.77%
capital K	6.82	6.59
output Y	2.00	1.97
wage w	1.28	1.26
labor tax τ	0	2.6%
private wealth A	6.82	7.77

Debt worth 1.18 units of goods entered; capital fell by only 0.23.



Of the 1.18 of new paper, households absorb 0.95 as extra wealth; capital gives up the other 0.23.

Partial Crowding Out

- Crowding-out coefficient: $\Delta K / \Delta B \approx -0.23/1.18 \approx \mathbf{-0.20}$. Twenty cents of capital per dollar of debt.
- Why so little? The household demand curve is steep near the equilibrium: small rises in r elicit large increases in desired buffers, so households absorb most of the paper.
- Contrast the benchmarks: representative-agent infinite-horizon steady state has $r = 1/\beta - 1$ pinned by preferences $\Rightarrow$ **zero** steady-state crowding out (Ricardian); OLG models without insurance motives can approach **one for one**.
- The Aiyagari economy sits in between, and *where* it sits is a quantitative property of the calibrated earnings risk.

The Welfare Trade-off

What moved, in welfare-relevant terms:

- **Benefit:** more of the safe asset at a higher return, so self-insurance is cheaper; constrained and low-wealth households value the liquidity function of public debt.
- **Costs:** w down 1.2% and τ up 2.6 points; and K down means lower long-run consumption capacity.
- Incidence is asymmetric by construction: the return gains accrue to asset holders; the wage and tax losses to workers. *Debt redistributes even in steady state.*
- Which force wins, and at what debt level? That is Aiyagari–McGrattan's question.

Debt as Public Liquidity: the General Point

- In incomplete-markets economies the private supply of safe stores of value is, in a precise sense, *too small*: $r^* < 1/\beta - 1$ signals scarcity of self-insurance vehicles.
- Government debt is a technology for manufacturing them: it converts future taxes into present safe assets.

What the Steady-State Comparison Cannot Tell Us

- Our table compares two *steady states*: economies that have always had $B/Y = 0$ or 0.6.
- An actual policy change travels a *path*: issuing the debt, changing taxes, capital decumulating slowly, prices moving along the way.
- Along the path, someone receives the proceeds and someone pays—generational and distributional incidence lives almost entirely in the transition.
- First: the optimal quantity of debt, steady state by steady state.

The Question

- Aiyagari and McGrattan (1998): choose the steady-state debt level B/Y to maximize (utilitarian) welfare in a calibrated incomplete-markets economy.
- The trade-off is exactly as before: debt provides self-insurance liquidity and pushes r toward the households' preferred rate—against tax distortions, a lower wage, and crowded-out capital.
- An interior optimum is plausible *a priori*; the contribution is the quantitative answer.

Their Model, Relative to Our Experiment

Three enrichments beyond the previous setup:

- **Endogenous labor supply** The labor tax distorts hours.
- **Transfers and government consumption** at calibrated U.S. levels, so the tax base and rates are realistic.
- Calibration to postwar U.S. aggregates and the earnings process of its era; the observed U.S. debt ratio ($\approx 2/3$ then) is *not* targeted—it is the object being evaluated.

The Headline

- Optimal steady-state debt: **about** 2/3 **of GDP**— very close to the U.S. level at that time.
- But the welfare function over B/Y is **extremely flat**: moving from the optimum to substantially different debt levels costs on the order of hundredths of a percent of consumption.
- Read both findings together: the liquidity benefit and the distortion cost nearly offset over a wide range.

Mechanism Accounting

Decomposing the near-offset at the margin:

- Raising B: r up (savers gain; self-insurance cheaper), w down and τ up (workers lose), K down (future consumption capacity down).
- The liquidity gain is concentrated where marginal utility is high—low-wealth households—but those same households earn mostly labor income, so they also bear the wage and tax costs.
- The internal offset *within* the constrained group is why the aggregate welfare function is flat: debt gives with the asset hand and takes with the labor hand from the same people.

Why Flatness Is a Finding, Not a Failure

- A flat objective means the *level* of debt is a second-order policy margin in this class of models—a genuinely useful negative result for debates that treat debt ratios as first-order welfare objects.
- It also means the sign and location of the optimum are **fragile**: modest changes in the insurance environment or the wealth distribution can move the optimum a lot without changing welfare much.

The Later Literature Moved the Answer

- Richer wealth distributions and MPC heterogeneity (matching the concentrated U.S. distribution) can flip the optimum toward *negative* debt—the government as a saver.
- Accounting for the *transition* to a new debt level (rather than comparing steady states) changes rankings again: reducing debt requires a painful high-tax traverse that steady-state comparisons ignore.

Assessment

What to retain from Aiyagari–McGrattan:

- The right *language* for debt policy in HA economies: liquidity provision vs. distortion and crowding out, priced in one general equilibrium.
- The methodological template: calibrate, compute steady states across policy values, trace the welfare function.
- And its declared limitation: steady-state welfare comparison. Two households with the same steady-state welfare can fare very differently *getting there*.

Which Raises the Method Question

- Every claim so far compared long-run equilibria.
- Policy evaluation needs the **path**: prices, quantities, and the distribution, period by period, from the old equilibrium to the new one.

Why Transitions Change Answers

Two distinct reasons:

- **Aggregate:** capital moves slowly. A reform whose new steady state has higher consumption may require a long low-consumption traverse (accumulating the capital); with discounting, the traverse can outweigh the destination.
- **Distributional:** the path reprices assets and wages *now*. Whoever holds wealth when r jumps, or works when w falls, bears gains and losses that no steady-state comparison records.
- Domeij–Heathcote is the canonical demonstration that both effects can *reverse* a steady-state welfare ranking.

The Object We Compute

A **perfect-foresight transition** ("MIT shock"):

- At $t = 0$ the economy sits in the initial stationary equilibrium; an unanticipated, permanent policy change is announced and implemented.
- From $t = 0$ on, everyone has perfect foresight over the entire path of prices and policies.
- We seek sequences $\{r_t, w_t, \tau_t, K_t, \lambda_t\}_{t=0}^{T}$, with T large enough that the economy has effectively arrived at the new stationary equilibrium.
- Nonstationary problem $\Rightarrow$ time-indexed value functions—the same structure as Ben-Porath's finite horizon, now with a distribution attached.

Warm-up: the Shooting Algorithm (RA Model)

In the representative-agent growth model (L&S, RMT 2nd ed., §11.3.3; transition experiments in §11.6), the transition solves a two-point boundary value problem:

- Given K_0, guess C_0; iterate the Euler equation and resource constraint forward; adjust C_0 until the path converges to the new steady state rather than exploding— **shooting**.
- One unknown number, one terminal condition: the saddle path.
- The HA analogue cannot shoot on one number: the "initial condition" includes an entire *distribution*, and household behavior depends on the whole future price path. Hence a fixed-point-in-sequences method.

The HA Transition Algorithm: Overview

1. Compute the **initial** and **final** stationary equilibria.
2. **Guess a path** for the market-clearing object, e.g. $\{K_t\}_{t=1}^{T}$ (equivalently $\{r_t, w_t\}$), ending at the final steady state.
3. **Backward pass:** from the final steady-state value function, solve the time-indexed Bellman equations $V_T, V_{T-1}, \ldots, V_0$ under the guessed prices.
4. **Forward pass:** starting from the *initial* distribution, push λ_t forward with the time-t policies; aggregate to get implied $\{\hat{K}_t\}$.
5. **Update** the guess (with dampening) until implied and guessed paths coincide.

Step 1: Endpoints

- Initial stationary equilibrium: the pre-reform economy; its distribution λ_0 is the transition's initial condition.
- Final stationary equilibrium: the post-reform economy; its value function is the transition's *terminal* condition.
- Note the asymmetry, and memorize it: **distributions travel forward from the past; values travel backward from the future.** The algorithm is built around that crossing.
- Sanity check before proceeding: both endpoints solved to tight tolerance—transition errors inherit endpoint errors.

Steps 2–3: Prices and the Backward Pass

- A convenient initial guess: interpolate K_t smoothly (linearly, or by the RA model's path) between K_{init} and K_{final}.
- Prices along the guess: $r_t = \theta(K_t/L_t)^{\theta-1} - \delta$, $w_t = (1-\theta)(K_t/L_t)^{\theta}$; fiscal variables from the policy rule (e.g. τ_t balancing the budget given B_t).
- Backward pass: set $V_T = V_{\text{final SS}}$; for $t = T-1, \ldots, 0$ solve one Bellman step at prices (r_t, w_t, τ_t), storing the *policy* $g_{a,t}$.
- Cost: T single Bellman steps—no fixed points inside; EGM makes each step cheap.

Steps 4–5: the Forward Pass and the Update

- Forward pass: $\lambda_{t+1} = \Gamma(g_{a,t})\,\lambda_t$ starting from λ_0; aggregate assets each period; subtract B_t to get implied capital $\hat{K}_{t+1}$.
- Update: $K_t^{new} = \xi\,\hat{K}_t + (1-\xi)\,K_t$ with dampening $\xi \in (0.1, 0.5)$; undamped updates oscillate.
- Convergence metric: $\max_t |\hat{K}_t - K_t|$ below tolerance *and* the tail of the path flat at the final steady state.
- If the path still moves at $t = T$: T is too short. Rule of thumb for annual calibrations: T of 150–300; check by doubling.

Welfare Along the Path

- The object that justifies the machinery: $V_0(a, e)$ from the backward pass is each household's value of the reform, *inclusive of the entire transition*.
- Compare with the pre-reform value function to get consumption-equivalent gains/losses **household by household**: a full distributional incidence map, not one aggregate number.
- Aggregation is then a choice you must defend: utilitarian, median-voter (who wins a referendum?), or Pareto-weight experiments.

The Experiment

- Policy question: **eliminate capital income taxation**, replacing the revenue with labor (or consumption) taxes.
- Complete-markets intuition says: do it. Capital taxes distort the intertemporal margin and steady-state comparisons show sizable aggregate gains (K, Y, w all up).
- Domeij and Heathcote re-run the experiment with (by now) familiar ingredients: incomplete markets, borrowing constraints, *and the full transition*.

What the Transition Does to the Reform

The path, in words:

- On impact: capital-tax revenue disappears, so the labor tax **jumps**; the after-tax wage falls immediately.
- Capital accumulates only *slowly*; the higher w and output that justify the reform arrive years later.
- Meanwhile the return to existing wealth rises immediately—a windfall to current asset holders on the stock they already own.
- So the reform front-loads costs onto labor income and front-loads gains onto wealth.

The Headline

- In the calibrated economy, accounting for the transition and heterogeneity, **a majority of households lose** from eliminating capital taxation—even though the new steady state looks better on aggregates.
- The winners are concentrated at the top of the wealth distribution: large asset positions, little reliance on the (temporarily heavily taxed) wage.
- The wealth-poor and constrained—high MPCs, labor-income dependent—bear the traverse and discount the distant wage gains.
- Steady-state welfare comparisons of this reform are not mildly optimistic; they get *the sign of the majority's welfare change wrong*.

Why This Paper Is on the Syllabus

- Methodologically: it is the cleanest published execution of exactly the transition dynamics algorithm.
- Substantively: it establishes the course's fiscal-policy hygiene rule—**no welfare claim about a reform without the transition and the distribution**.
- Politically: the incidence map explains why efficient-in-the-long-run reforms fail referenda; the median voter lives on the transition path, not in the steady state.

Caveats and Frontiers

- Compensation: could transfers along the path convert the reform into a Pareto improvement? Sometimes—but the compensating transfers must themselves be financed with distortionary taxes; the answer is quantitative, not free.
- The reform mix matters: replacing capital taxes with *consumption* taxes (which partly fall on existing wealth) changes incidence substantially.
- Anticipation matters: pre-announced reforms trigger accumulation before implementation.
- All three are viable extensions for the replication project.